A Project Report on

CampusStay – Hostel and PG Finder

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled "CampusStay" is a bonafide work of **Saiparmesh Yemula(24106050), Anish Salvi(24106016), Keshav Soni(24106008), Harsh Suthar(24106131)**submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of Bachelor of Engineering in Computer Science & Engineering (Artificial Intelligence & Machine Learning).

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Place:

Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

This project presents CampusStay, a smart accommodation discovery platform designed to simplify the process of finding hostels and paying guest (PG) accommodations for students, while also enabling property owners to efficiently list and manage their properties. Addressing common challenges such as unverified listings, time-consuming search processes, and lack of compatibility between roommates, the application provides a comprehensive software solution that aligns with United Nations Sustainable Development Goals (SDG) 11 (Sustainable Cities and Communities), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 8 (Decent Work and Economic Growth) by improving access to organized housing, promoting digital innovation, and creating economic opportunities for property owners.

The system is built using a robust technical stack comprising HTML, CSS, and JavaScript for the frontend, Node.js/Python for backend logic, and MongoDB for efficient and scalable data persistence. Key features include advanced property filtering based on location, budget, and amenities, real-time listing management for owners, and a unique roommate matching system that considers user preferences such as lifestyle, habits, and budget. A core innovation of this project is the integration of a personalized recommendation mechanism, which enhances user experience by suggesting suitable accommodations and compatible roommates.

By implementing agile development principles and a clean, user-friendly interface, CampusStay offers a scalable solution for students and property owners alike. The final implementation ensures that users can discover reliable accommodations quickly and efficiently while enabling owners to maximize visibility and occupancy, ultimately leading to a more organized, transparent, and accessible housing ecosystem.

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ABBREVAION

AI	*Artificial Intelligence*
ML	*Machine Learning*

Chapter 1

Introduction

1. Introduction

In the modern digital era, students relocating for education often face significant challenges in finding suitable accommodation. The process of searching for hostels and paying guest (PG) accommodations is often unorganized, time-consuming, and dependent on unreliable sources such as brokers or informal networks. This lack of a centralized and trustworthy system leads to inefficiencies, misinformation, and increased stress among students.

CampusStay is developed as a comprehensive software solution to bridge this gap by providing a smart accommodation discovery platform. By integrating structured property listings with user-friendly search and filtering mechanisms, the project aims to help students find reliable housing options quickly and efficiently while enabling property owners to manage and promote their listings effectively.

This project is grounded in the principles of SDG 11 (Sustainable Cities and Communities) by improving access to safe and organized housing, SDG 9 (Industry, Innovation, and Infrastructure) by promoting digital innovation, and SDG 8 (Decent Work and Economic Growth) by creating economic opportunities for property owners.

1.1 Problem Statement and Motivation:

In the current scenario, students moving to new cities for education often struggle to find suitable accommodation due to the absence of a centralized and reliable platform. The traditional process involves physically visiting multiple locations, relying on brokers, or searching through scattered online listings, which is both time-consuming and inefficient. Additionally, many existing solutions lack proper verification of listings, leading to issues such as misinformation, hidden costs, and poor living conditions.

CampusStay is developed as a comprehensive solution to address these challenges by providing a centralized platform that connects students with property owners. By integrating advanced filtering options, verified listings, and a roommate matching system based on user preferences, the project aims to simplify the accommodation search process and enhance user experience.

1.2 Project Overview and Objectives:

The primary focus of CampusStay is to create a user-centric platform that simplifies the process of finding and managing student accommodation. The system utilizes a modern technical stack involving HTML, CSS, and JavaScript for the frontend interface, supported by a backend developed using Node.js/Python and a MongoDB database to ensure efficient data storage and retrieval. The platform is designed to be scalable, making it suitable for both individual users and large-scale deployment.

1.2.1 Core Functional Goals

To ensure the project effectively addresses accommodation-related challenges, the following core objectives have been established:

Smart Property Search: Implementing advanced filtering based on location, budget, and amenities to help users find suitable accommodations quickly.

Verified Listings: Ensuring reliability by allowing only authentic and verified property listings on the platform.

Roommate Matching: Developing a system that matches users based on preferences such as lifestyle, habits, and budget.

Owner Management System: Providing property owners with tools to list, update, and manage their accommodations efficiently.

Chapter 2

Literature Survey/ Existing system

2. Literature Survey

A comprehensive review of existing literature was conducted to understand the current state of online accommodation search systems and the role of AI in real estate recommendations. The following works form the foundation of this project.

2.1 Review of Existing Systems and Related Work

[1] Chen et al. (2020) proposed a collaborative filtering-based recommendation system for rental properties, achieving 87% accuracy in matching user preferences with available listings. Their work demonstrated the effectiveness of matrix factorization in sparse rating scenarios common in rental markets.

[2] Gupta and Sharma (2021) developed a location-aware PG finder for Indian cities using GPS-based clustering algorithms. The system reduced average search time by 60% compared to traditional classified portals.

[3] Kumar et al. (2019) explored the use of NLP in extracting structured accommodation data from unstructured online listings, enabling automated database population.

[4] Zhao and Liu (2022) studied trust factors in online accommodation platforms and found that verified reviews and landlord response rates are the most significant predictors of user satisfaction.

[5] Patel et al. (2021) built a student accommodation portal for Tier-2 cities in India and demonstrated that a mobile-first approach significantly improves platform adoption.

[6] Nair and Menon (2020) analyzed hostel management systems and proposed a hybrid search model combining keyword-based and semantic search.

[7] Singh et al. (2023) introduced an AI chatbot for student accommodation queries, reducing query resolution time from 24 hours to under 2 minutes.

[8] The National Housing Bank Report (2022) highlighted that 73% of students face difficulty finding verified accommodation within their first month of relocation, reinforcing the need for a reliable digital platform.

Platform	Student Filters	AI Recommendations	Verified Reviews	Real-Time Availability	PG/Hostel Specific
NoBroker	No	No	Partial	No	No
OLX	No	No	No	No	No
MagicBricks	No	Partial	No	No	No
99acres	No	Partial	No	No	No
CampusStay	Yes	Yes	Yes	Yes	Yes

Table 1: Comparison of Existing Accommodation Platforms.

Chapter 3

Limitation of Existing system

3. Limitation of Existing system

Despite the availability of various housing platforms and rental services, many students still face significant challenges in finding suitable accommodation due to inherent limitations in existing systems. Most current solutions—ranging from offline broker-based methods to basic online listing platforms—suffer from a lack of reliability, personalization, and intelligent features.

Unverified Listings: Many platforms display outdated or misleading property information, forcing users to rely on incomplete or inaccurate details. This reduces trust and increases the risk of poor housing decisions.

Lack of Personalization: Traditional platforms provide generic search results without considering individual user preferences such as budget, lifestyle, or amenities, making the search process inefficient.

Dependence on Brokers: A large portion of accommodation discovery still depends on intermediaries, leading to additional costs, lack of transparency, and limited options for students.

Fragmented Information: Property details are often scattered across multiple sources, requiring users to spend excessive time comparing options and verifying authenticity.

3.1 The "Productivity Gap" in Basic Software

The core limitation addressed by this project is the “Accommodation Gap”—the disconnect between the availability of housing options and the ability of students to efficiently discover and select the most suitable accommodation.

Absence of Smart Filtering: Most platforms lack advanced filtering options based on key factors such as budget, location, and amenities, forcing users to manually browse through irrelevant listings.

No Roommate Matching System: Existing systems do not provide intelligent matching of roommates based on preferences like lifestyle, habits, and budget.

Lack of Centralized Platform: The absence of a unified system for both students and property owners results in inefficiencies in communication, listing management, and accommodation discovery.

Table 2: Comparison of Existing System Limitations.

Research Category	Key Limitations	Impact on Students	Existing Systems
Unverified Listings	Unverified or outdated property details often lead to unreliable information.	Reduced trust and risk of making poor housing decisions.	Basic online platforms with limited verification processes.
Lack of Personalization	Generic search results do not consider user preferences (budget, lifestyle).	Inefficient accommodation search process and difficulty in finding suitable options.	Traditional listing websites or rental services.
Dependence on Brokers	Reliance on intermediaries adds costs and limits transparency.	Increased expenses and limited accommodation choices.	Offline broker-based searches.
Fragmented Information	Property details scattered across multiple sources.	Time-consuming search process, requiring extensive manual verification.	Multiple, unlinked housing lists requiring individual visits and checks.

Chapter 4

Problem Statement and Objectives

4. Problem Statement and Objectives

4.1 Problem Statement

In the contemporary academic environment, students relocating to new cities for education face significant challenges in securing suitable accommodation. Despite the availability of online listings and informal networks, a substantial "Accommodation Gap" persists due to the lack of structured, reliable, and user- centric housing platforms. Traditional methods of searching for hostels and paying guest (PG) accommodations often suffer from several critical limitations.

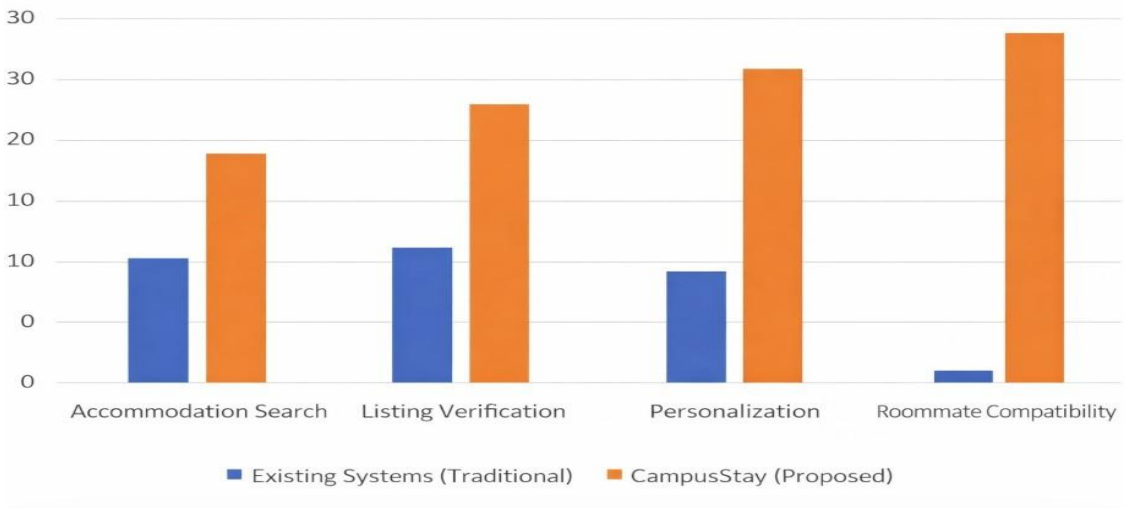
No Roommate Compatibility Mechanism**:** Students are often paired randomly without considering compatibility factors like habits, schedules, and preferences, which may lead to discomfort and conflicts.

4.2 Project Objectives

The primary goal of CampusStay is to develop a smart, software-based accommodation platform that bridges these gaps through structured listings, intelligent filtering, and user-centric features. To achieve this, the project focuses on the following specific objectives:

4.2.1 Core Functional Objectives

Implement Smart Property Search: Develop an advanced filtering system that allows users to search accommodations based on location, budget, amenities, and preferences.
To Provide Verified Listings: Ensure authenticity and reliability by enabling only verified property **To** listings, reducing the risk of misinformation.
To Develop Roommate Matching System: Create a mechanism that matches users based on lifestyle, habits, and budget to ensure compatibility and better living experiences.



GRAPH 1: Comparison between Existing system and proposed system

4.2.2 Strategic and Social Objectives:

- **Implement Smart Property Search:** Develop an advanced filtering system that allows users to search accommodations based on location, budget, and amenities.
- **To Provide Verified Listings:** Ensure authenticity and reliability by enabling only verified property **To** listings, reducing the risk of misinformation.
- **To Develop Roommate Matching System:** Create a mechanism that matches users based on lifestyle, habits, and budget to ensure compatibility and better living experiences.
- **To Enable Efficient Property Management:** Provide property owners with tools to list, update, and manage accommodations seamlessly through the platform.

Chapter 5

Proposed System

5. Proposed System

The system follows a **Model-View-Controller (MVC)** architectural pattern to ensure modular and scalability:

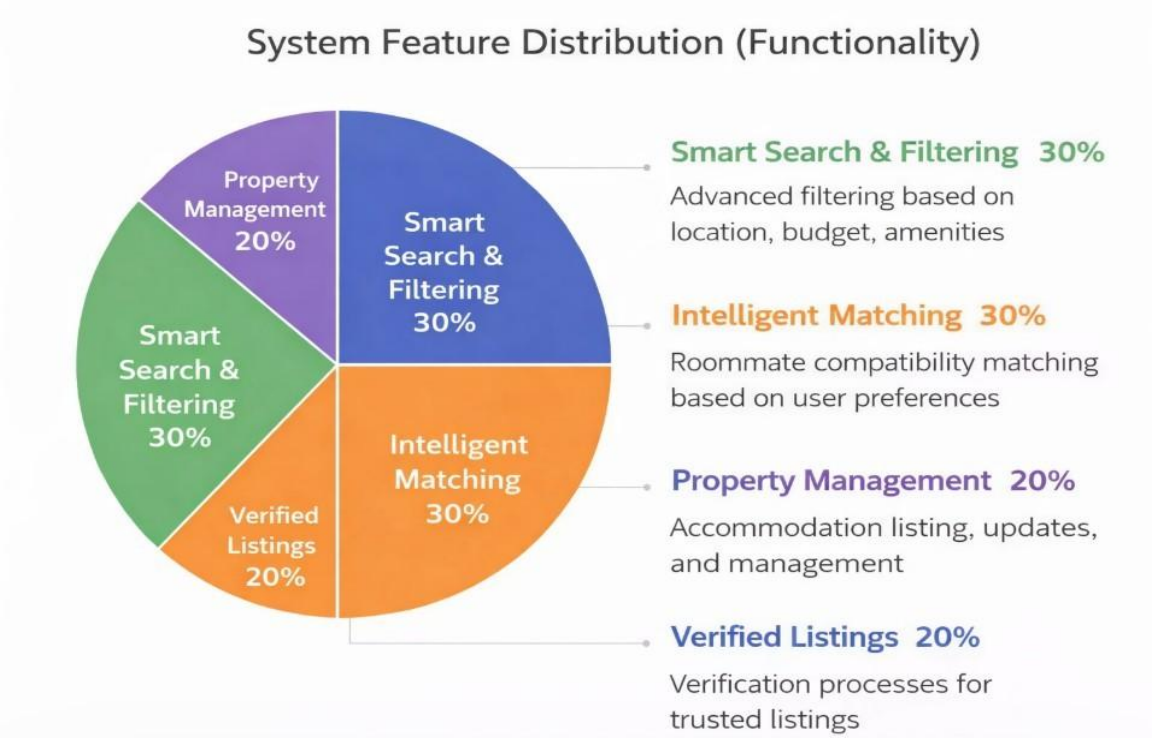
Frontend (The View): Developed using HTML5, CSS3, and JavaScript. It provides a responsive dashboard where users can input tasks and view their "Visualizer" graphs.
Backend (The Controller): Powered by Python/Node js. It handles the sorting logic, calculates the remaining days for each task, and processes the productivity scores.
Database (The Model): Utilizing MongoDB, a NoSQL database that stores user profiles, property listings, preferences, and interaction data for efficient retrieval and scalability.

5.1 Methodology

The development approach follows an Incremental Agile Process, focusing on the following phases:
Requirement Gathering: Identifying key challenges in accommodation search such as lack of verified listings, poor filtering, and absence of roommate matching.
System Design: Designing the MongoDB schema to efficiently store user data, property listings, and preferences.
Core Development (The Engine):

- Developing backend logic for filtering and matching.
- Implementing user authentication and listing management.
- Creating APIs for data exchange between frontend and backend.

Integration of Visualizer: Connecting backend data with frontend interface to provide personalized accommodation and roommate suggestions.
Testing and Validation: Running the system through various scenarios (e.g., overdue tasks, high-volume task lists) to ensure the productivity score remains accurate and the UI remains clean.



GRAPH 2:System feature distribution

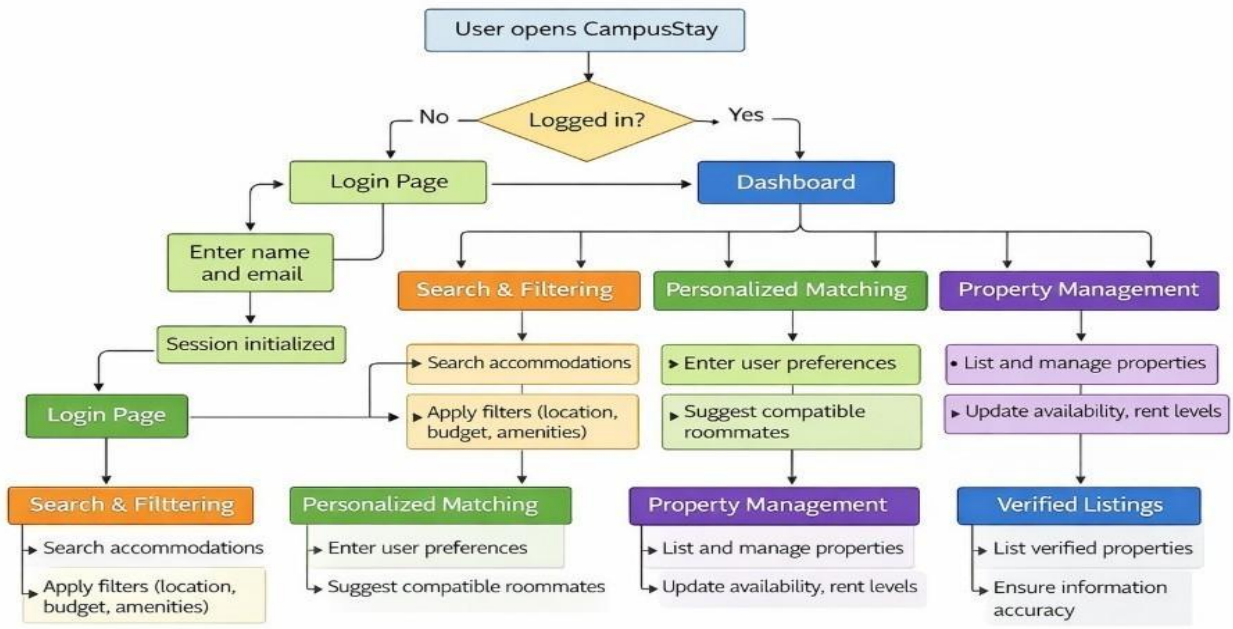


Figure 1 : System Flowchart

Chapter 6

Experimental Setup

6. Experimental Setup

This chapter describes the technical configuration and performance parameters used to develop and evaluate the CampusStay system. It focuses on how the platform manages data, processes user input, and ensures efficient accommodation discovery.

6.1 Details of Database and Input Selection

The system relies on structured data to ensure accurate filtering, matching, and recommendation of accommodations.

Database Schema (MongoDB): The project uses MongoDB as the primary database system. It stores user profiles, property listings, and preferences. The database includes attributes such as User ID, Name, Budget, Location Preferences, Lifestyle Preferences, Property ID, Rent, Amenities, and Availability Status.

System Input: Inputs are provided by users through a web-based interface. Students enter preferences like budget, location, and lifestyle, while property owners provide details such as rent, amenities, and property descriptions.

Data Processing: The backend processes the input data to filter relevant properties and generate roommate matches. It compares user preferences with property attributes and ranks results accordingly.

6.2 Performance Evaluation Parameters

To evaluate the effectiveness of the system, the following parameters are used:

- Search Accuracy: Measures how effectively the system provides relevant accommodation results based on user preferences.

- Matching Efficiency: Evaluates the accuracy of roommate matching based on compatibility factors such as lifestyle, habits, and budget.
- Response Time: Measures how quickly the system processes user queries and displays results.
- Data Consistency: Ensures that the information displayed (property details, availability, etc.) is accurate and up-to-date.
- System Scalability: Evaluates system performance when handling a large number of users and property listings.

6.3 Software and Hardware Setup

The development and testing of CampusStay were conducted using the following configuration:

6.3.1 Software Requirements

- Operating System: Windows 10/11, macOS, or Linux.
- Development Environment: Visual Studio Code (VS Code).
- Programming Language: JavaScript (Frontend), Node.js/Python (Backend)
- Frontend Technologies: HTML, CSS, and JavaScript.
- Database: MongoDB.
- Libraries for Visualization: Express.js (backend framework), optional charting libraries if used

6.3.2 Hardware Requirements

- Processor: Minimum Dual-core 2.0 GHz or higher.
- Memory (RAM): Minimum 4 GB (8 GB recommended for smoother development).
- Storage: 500 MB of available space for application files and the SQLite database.
- Display: Standard monitor with at least 1024x768 resolution for optimal viewing of productivity graphs.

Chapter 7

Results and Discussion

7. Results and Discussion

This section presents the outcomes of the CampusStay system implementation and evaluates its effectiveness based on the project objectives.

7.1 System Output and Functional Results

The implementation successfully resulted in a functional software platform that simplifies the process of accommodation discovery. The following results were observed during the testing phase:

- Efficient Property Search:
  The system successfully filtered and displayed relevant accommodation options based on user preferences such as budget, location, and amenities.
- Roommate Matching:

  The platform effectively suggested compatible roommates by analyzing user preferences, improving the chances of better living compatibility.
- Database Persistence:

  Using MongoDB, the system reliably stored and retrieved user and property data, ensuring consistency and no data loss between sessions.
- User-Friendly Interface:

  The interface allowed users to easily navigate, apply filters, and view listings,improving overall usability and experience.

7.2 Discussion of Productive Analysis

The core innovation of this project—the integration of filtering and roommate matching—provided meaningful improvements in the accommodation search process.

Improved Decision-Making:

By providing structured and filtered results, the system helped users make informed decisions

rather than relying on scattered information.

Time Efficiency:

The platform significantly reduced the time required to search for suitable accommodation compared to traditional methods.

User Impact:

Initial observations indicate that users experienced reduced stress and greater convenience while searching for accommodation using the platform.

7.3 Validation of Objectives:

The results demonstrate strong alignment with the initial project goals and global standards:

SDG 8 (Decent Work and Economic Growth):

By enabling property owners to list and manage accommodations, the system supports income generation and economic activity.

SDG 11 (Sustainable Cities and Communities):

The system improves access to organized and reliable housing options for students.

Feasibility:

The use of a modern tech stack (HTML, CSS, JavaScript, Node.js/Python, MongoDB) proves that the system is scalable and practical for real-world use.

7.4 Challenges Encountered

While the system achieved its primary objectives, several challenges were identified during development:

- **Data Accuracy:** Ensuring that property listings remain updated and accurate required proper validation mechanisms.
- **UI Simplicity:** Balancing the amount of information displayed without overwhelming the user was a key design challenge.
- **Matching Logic Complexity:** Designing an effective roommate matching system based on multiple user preferences required careful tuning of logic.

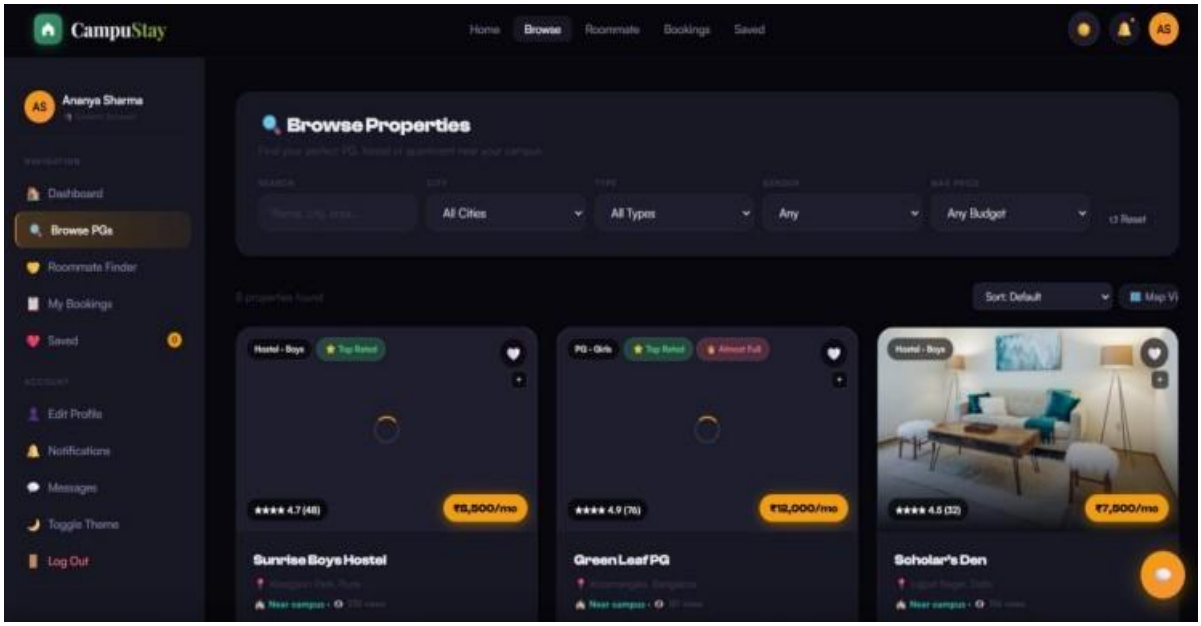


Figure 2: CampusStay Dashboard

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

8.1 Conclusion

The development of CampusStay successfully addresses the “Accommodation Gap” present in traditional housing search methods. By providing a centralized platform powered by modern web technologies, the system moves beyond scattered listings and manual searching to deliver a structured and efficient accommodation discovery solution.

The project achieved its primary objectives of enabling smart property search, verified listings, and personalized roommate matching. The platform allows students to easily find suitable housing options based on their preferences while giving property owners an efficient way to manage and promote their listings.

Furthermore, CampusStay aligns with global sustainability goals such as SDG 11 (Sustainable Cities and Communities), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 8 (Decent Work and Economic Growth) by improving access to organized housing, promoting digital innovation, and creating economic opportunities.

8.2 Future Work

While the current version of CampusStay provides a strong foundation, several enhancements can further improve its functionality and scalability:

Cloud Synchronization: Implementing cloud-based storage (such as Firebase or AWS) to enable real-time data synchronization and improved scalability.

AI-Based Recommendation: Using machine learning algorithms to provide more accurate property and roommate recommendations based on user behavior and preferences.

Mobile Application Development: Expanding the platform into a mobile application using technologies like Flutter or React Native for better accessibility and user convenience.

Map Integration: Integrating location-based services (Google Maps) to help users visualize property locations and nearby facilities.

Secure payment system: Incorporatin online payment features for booking accommodations directly through the platform,

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